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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Analog to Digital and Digital to Analog Converters

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

24	19	1990 – 2025	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	12 questions
Marks 2	12 questions
Answer key	at the end

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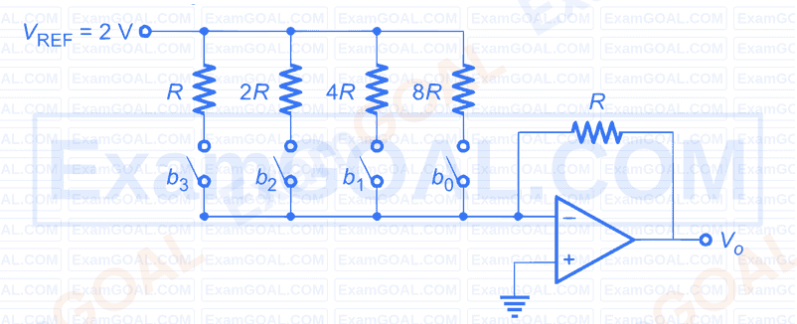
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Marks 1 12 questions

Q1. GATE ECE 2025

A 4-bit weighted-resistor DAC with inputs b_3, b_2, b_1, and b_0 (MSB to LSB) is designed using an ideal opamp, as shown below. The switches are closed when the corresponding input bits are logic ' 1 ' and open otherwise. When the input $b_3b_2b_1b_0$ changes from 1110 to 1101, the magnitude of the change in the output voltage V_O (in mV , rounded off to the nearest integer) is _____.



Open solution

Q2. GATE ECE 2023

The signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input is given to be 61.96 dB. The resolution of the ADC is _____ bits (rounded off to the nearest integer).

Open solution

Q3. GATE ECE 2020

A 10 bit D/A converter is calibrated over the full range 0 to 10 V . If the input to the D/A converter is 13 A (in hex), the output (rounded off to three decimal places) is ____ V.

Open solution

Q4. GATE ECE 2015 Set 1

Consider a four bit D to A converter. The analog value corresponding to digital signals of values 0000 and 0001 are 0 V and 0.0625 V respectively. The analog value (in Volts) corresponding to the digital signal 1111 is _____.

Open solution

Q5. GATE ECE 2002

The number of comparators required in a 3-bit comparator type ADC is

(A) 2 (B) 3 (C) 7 (D) 8

Open solution

Q6. GATE ECE 2000

The number of comparators in 4-bit flash ADC is

(A) 4 (B) 5 (C) 15 (D) 16

Open solution

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Q7. GATE ECE 1999

The resolution of a 4-bit counting ADC is 0.5 Volts. For an analog input of 6.6 Volts, the digital output of the ADC will be

(A) 1011 (B) 1101 (C) 1100 (D) 1110

Open solution

Q8. GATE ECE 1998

The advantage of using a dual slope ADC in a digital voltmeter is that

(A) its conversion time is small
(B) its accuracy is high
(C) it gives output in BCD format
(D) it does not require a comparator.

Open solution

Q9. GATE ECE 1998

For an ADC, match the following : if

List 1

A. Flash converter
B. Dual slope converter
C. Successive approximation Converter

List 2

1. requires a conversion time of the order of a few seconds
2. requires a digital- to- analog converter
3. minimizes the effect of power supply interference.
4. requires a very complex hardware.
5. It is a tracking A/D converter.

Open solution

Q10. GATE ECE 1996

A 12-bit ADC is operating with a 1μ sec clock period and the total conversion time is seen to be 14 μ sec. The ADC must br of the

(A) flash type
(B) counting type
(C) integrating type
(D) successive approximation type.

Open solution

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Q11. GATE ECE 1994

Open solution

Match the List-1 (type of 8-bit ADC) with List-2(Minimum conversion time in clock cycles)

List - 1

A. Successive approximation
B. Dual-slope
C. Parallel Comparator

List - 2

1) 1
2) 8
3) 16
4) 256
5) 512

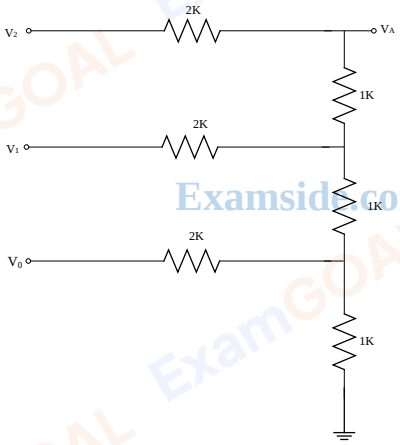
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Q12. GATE ECE 1990

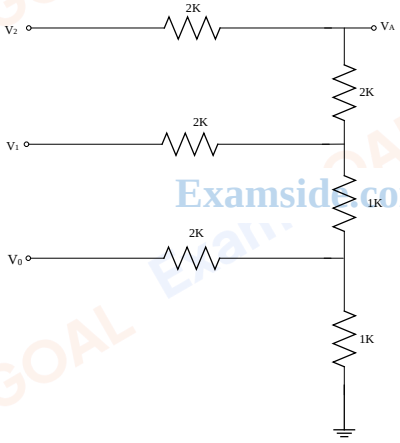
Which of the resistance networks of figure can be used as 3 bit R-2R ladder DAC. Assume V_0 corresponds to LSB.

Open solution

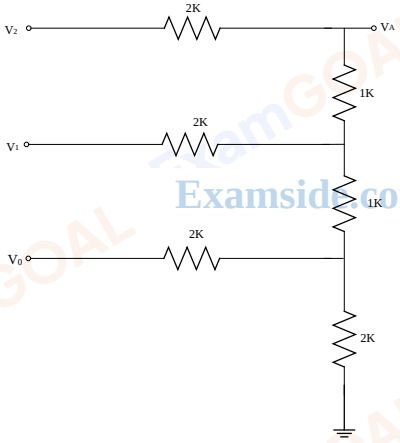
(i)



(ii)



(iii)



(A) Both (i) and (ii)

(B) Both (i) and (iii)

(C) Only (iii)

(D) Only (ii)

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Marks 2 12 questions

Q13. GATE ECE 2025

A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V . For an input sinusoidal signal with frequency 500 kHz , the maximum SNR (in dB, rounded off to two decimal places) and the data rate (in Mbps) at the output of the ADC are _____ , respectively.

Open solution

(A) 61.96 and 10 (B) 61.96 and 5 (C) 33.36 and 10 (D) 33.36 and 5

Q14. GATE ECE 2024

A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a normalized power of 1 W, and the total noise and distortion normalized power is 10 μW. The effective number of bits (rounded off to the nearest integer) of the ADC is _____.

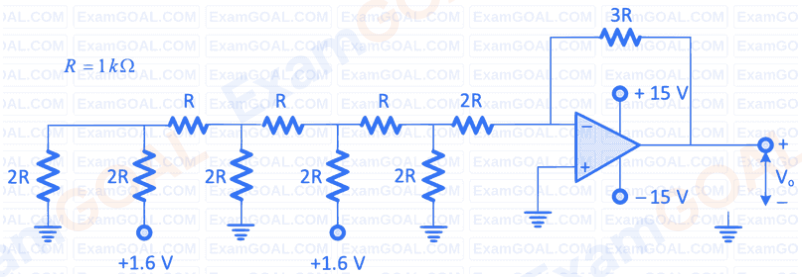
Open solution

(A) 7 (B) 8 (C) 9 (D) 10

Q15. GATE ECE 2022

Consider the circuit shown with an ideal OPAMP. The output voltage V_0 is _____ V (rounded off to two decimal places).

Open solution



Q16. GATE ECE 2021

An 8-bit unipolar (all analog output values are positive) digital-to-analog converter (DAC) has a full-scale voltage range from 0 V to 7.68 V . If the digital input code is 10010110 (the leftmost bit is MSB). Then the analog output voltage of the DAC (rounded off to one decimal place) is _____ V.

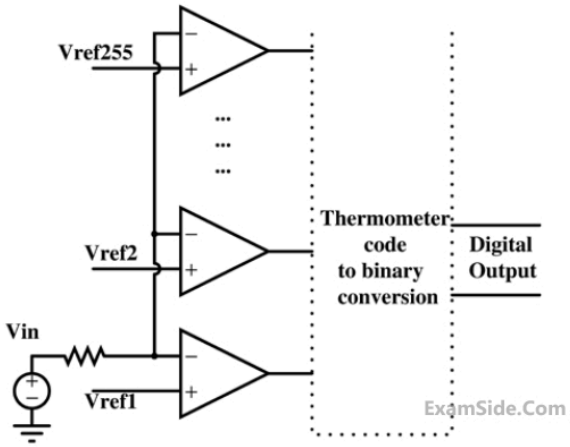
Open solution

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Q17. GATE ECE 2016 Set 2

Open solution

In an N bit flash ADC, the analog voltage is fed simultaneously to $2^N - 1$ comparators. The output of the comparators is then encoded to a binary format using digital circuits. Assume that the analog voltage source V_{in}(whose output is being converted to digital format) has a source resistance of 75 Ω as shown in the circuit diagram below and the input capacitance of each comparator is 8 pF. The input must settle to an accuracy of 1/2 LSB even for a full scale input change for properconversion. Assume that the time taken by the thermometer to binary encoder is negligible.



If the flash ADC has 8 bit resolution, which one of the following alternatives is closest to the maximum sampling rate?

(A) 1 megasamples per second

(B) 6 megasamples per second

(C) 64 megasamples per second

(D) 256 megasamples per second

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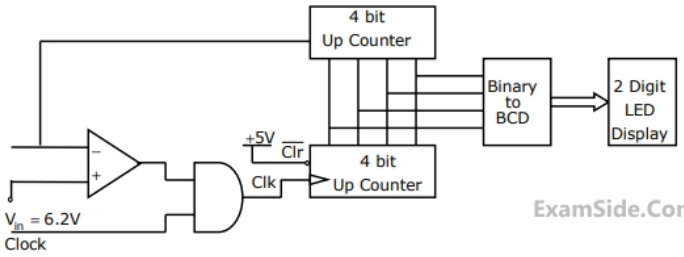
Q18. GATE ECE 2008

Open solution

In the following circuit, the comparator output is logic "I" if $V_1 > V_2$ and is logic "0" otherwise. The D/A conversion is done as per the relation

$$V_{DAC} = \sum_{n=0}^{3} 2^{n-1} b_n \; Volts,$$

where b_3(MSB), b_2, b_1 and b_0 (LSB) are the counter outputs.
The counter starts from the clear state.



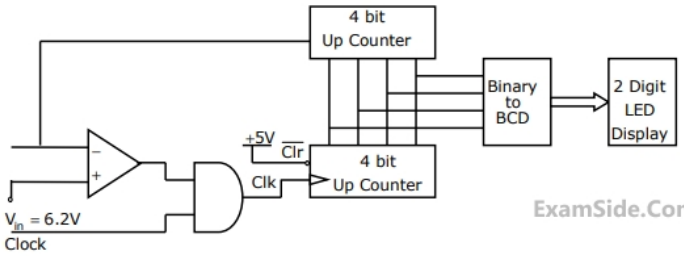
The stable reading of the LED display is

(A) 06 (B) 07 (C) 12 (D) 13

Q19. GATE ECE 2008

Open solution

In the following circuit, the comparator output is logic "1" if $V_1 > V_2$ and is logic "0" otherwise. The D/A conversion is done as per the relation $V_{DAC} = \sum_{n=0}^{3} 2^{n-1} b_n \; Volts$, where b_3(MSB), b_2, b_1 and b_0 (LSB) are the counter outputs.
The counter starts from the clear state.



The magnitude of the error between V_{DAC} and V_{in} at steady state in volts is

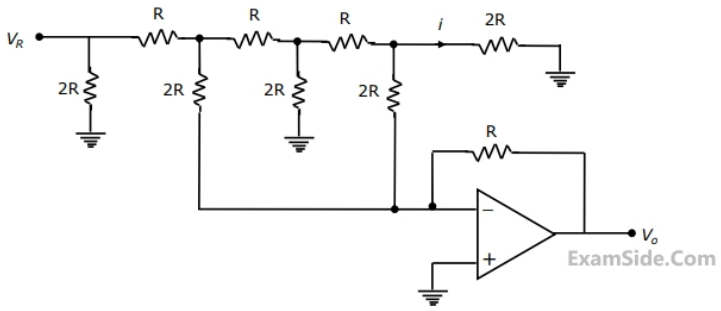
(A) 0.2 (B) 0.3 (C) 0.5 (D) 1

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Q20. GATE ECE 2007

Open solution

In the digital-to-Analog converter circuit shown in the figure below,
$V_R = 10V$ and $R = 10k\Omega$



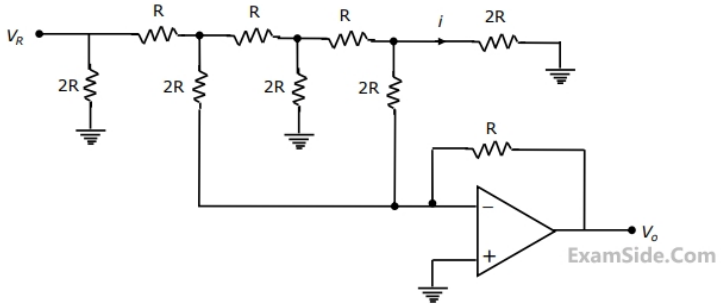
The current i is

(A) $31.25\,\mu A$ (B) $62.5\,\mu A$ (C) $125\,\mu A$ (D) $250\,\mu A$

Q21. GATE ECE 2007

Open solution

In the digital-to-Analog converter circuit shown in the figure below,
$V_R = 10V$ and $R = 10k\Omega$



The voltage V_0 is

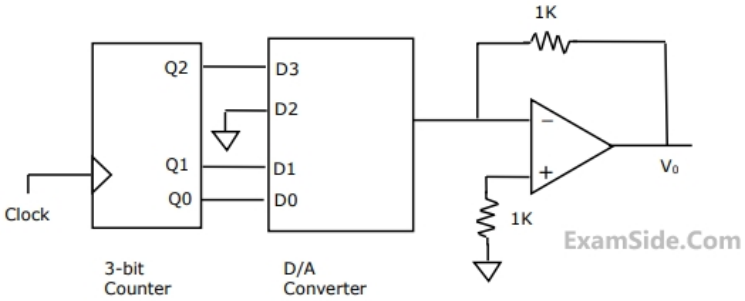
(A) -0.781V (B) -1.562V (C) -3.125V (D) -6.250V

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Q22. GATE ECE 2006

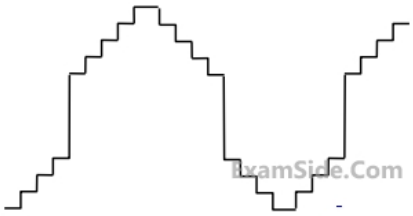
Open solution

A 4-bit D/A converter is connected to a free-running 3-bit UP counter, as shown in the following figure. Which of the following waveforms will be observed at V_0=?



In the figure shown above, the ground has been shown by the symbol ∇

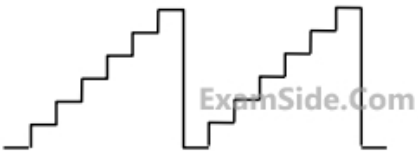
(A)



(B)



(C)



(D)

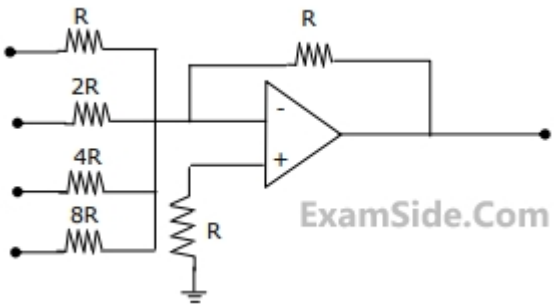


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Q23. GATE ECE 2003

Open solution

The circuit shown in figure is a 4-bit DAC



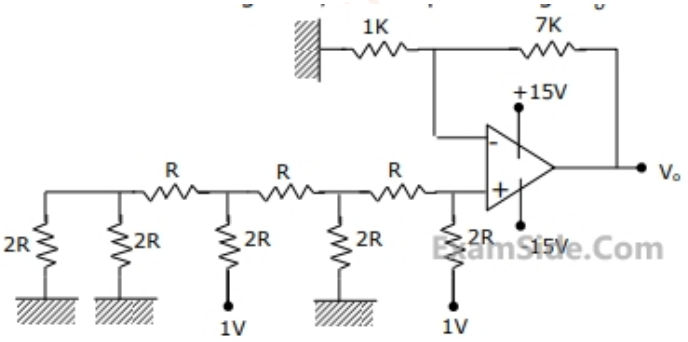
The input bits 0 and 1 are represented by 0 and 5 V respectively. The OP- AMP is ideal, but all the resistances and the 5V inputs have a tolerance of ±10%.
The specification (rounded to the nearest multiple of 5%) for the tolerance of the DAC is

(A) ± 35% (B) ± 20% (C) ± 10% (D) ± 5%

Q24. GATE ECE 2000

Open solution

For the 4 bit DAC shown in Figure, the output voltage V_0 is



(A) 10V (B) 5V (C) 4V (D) 8V

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Answer Key

Marks 1

1. **250**	2. **10**	3. **2.95 to 3.15**	4. **0.93**	5. **C**
6. **C**	7. **D**	8. **B**	9. **A-4, B-3, C-2**	10. **D**
11. **A- 2, B- 5, C- 1**	12. **C**			

Marks 2

13. **A**	14. **B**	15. **-0.5**	16. **4.5**	17. **A**
18. **D**	19. **B**	20. **B**	21. **C**	22. **B**
23. **A**	24. **B**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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